

IT Research Methodology

Quantitative Data Collection

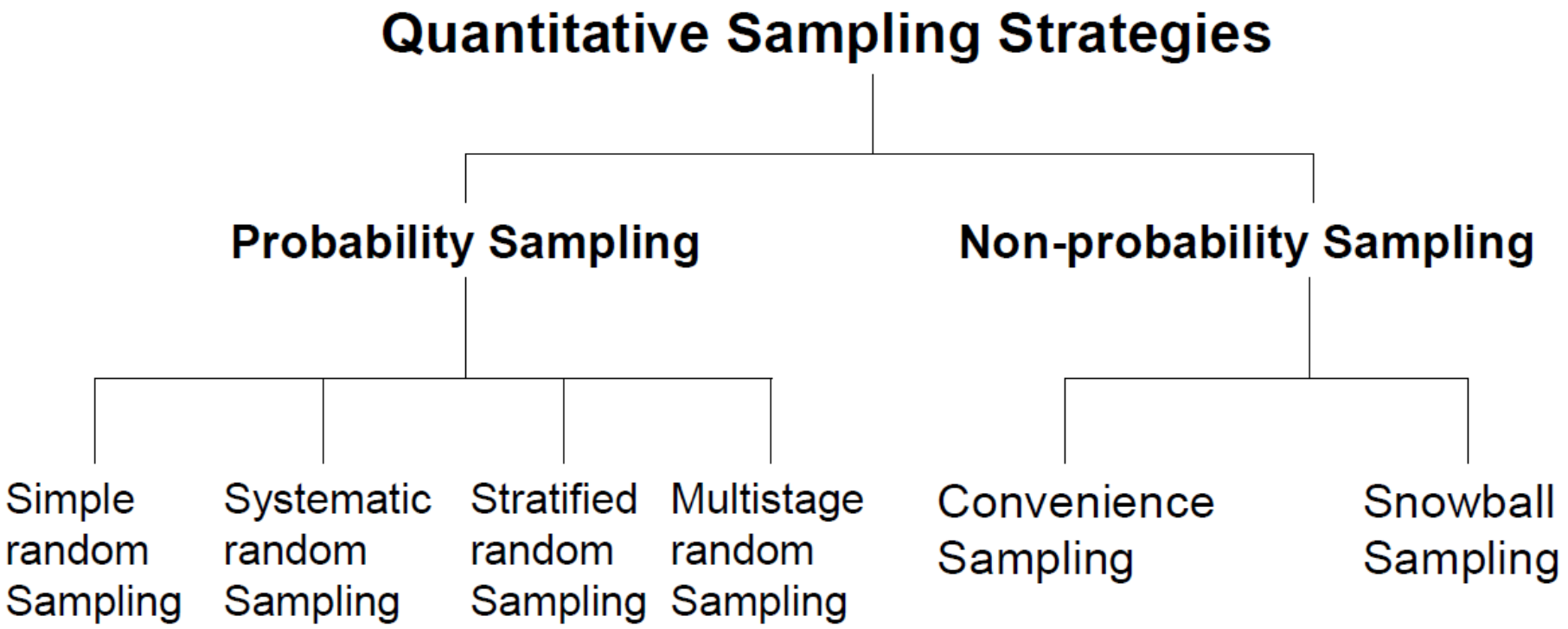
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Prepared by: CHEN SOVANN

Outline

- Type of Quantitative Sampling Strategies
- Sampling
- Samples and Populations
- Target and Accessible Population
- Representative Sample
- Sample Size
- Procedure of Data Collection
- Quantitative Sample Size
- Types of Instrument

❖ Type of Quantitative Sampling Strategies



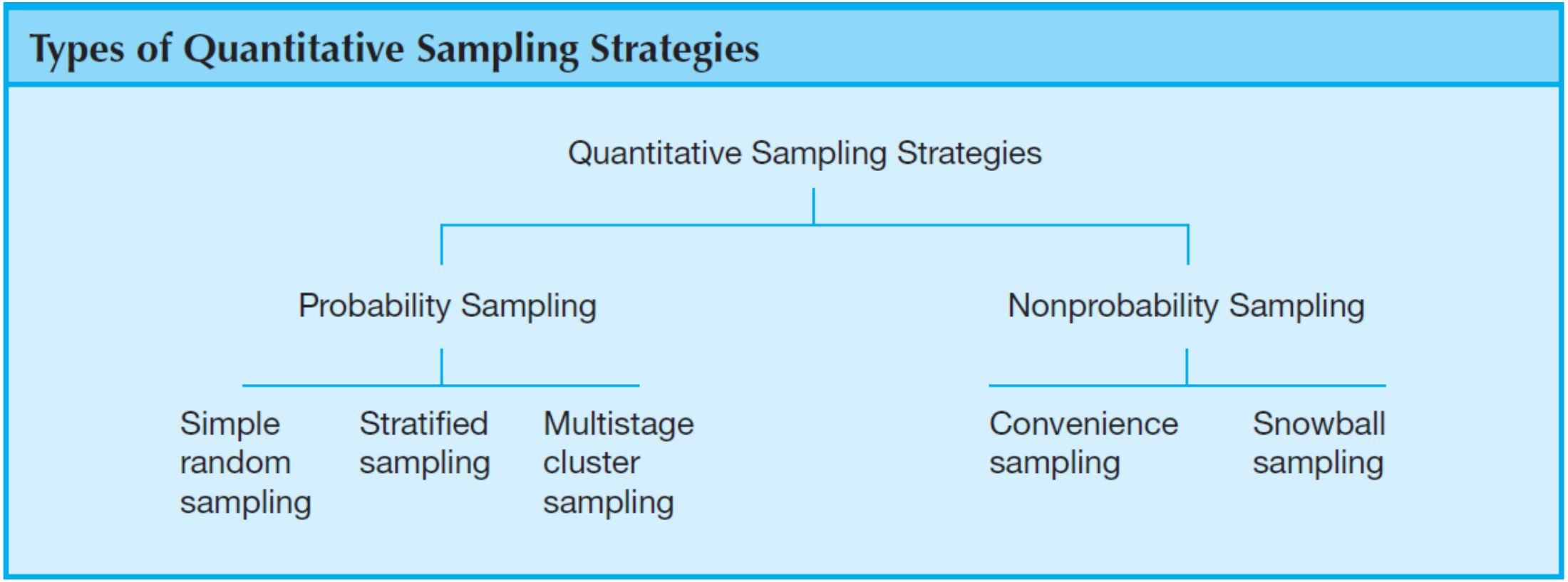
- ✓ The sample is representative of the population
- ✓ Make the generalizations to the population

- ✓ Convenience
- ✓ Represent some characteristic

❖ Sampling

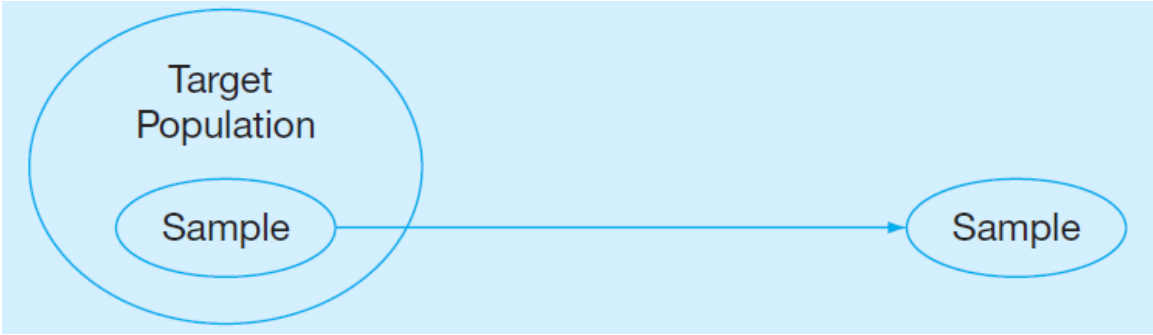
- **Sampling**: the process of selecting the individuals who will participate (e.g. be observed or questioned) in a research study.
- There are different methods of sampling (i.e., different ways of selecting individuals)
- The best sampling method depend on the research situation and the type of information that the researcher wishes to obtain.

❖ Sampling


Types of Quantitative Sampling Strategies
Quantitative Sampling Strategies
Probability Sampling
Nonprobability Sampling
Simple random sampling
Stratified sampling
Multistage cluster sampling
Convenience sampling
Snowball sampling

❖ Samples and Populations

- **Sample**: any part of population of individuals on whom information is obtained. It may, for a variety of reasons, be different from the sample originally selected.
- **Population**: all the member of the particular group. It is the group of interest to the researcher, the group of whom the researcher would like to generalize the result of the study.



❖ Target and Accessible Population

- **Target population**: the population of interest to the researchers. It is the population for whom we would like to answer the research question.
- **Accessible population**: the part of the target population that the researcher actually has access to. The researcher will draw the sample from accessible population.

❖ Representative Sample

- **Representative sample**: a sample that is like the population in terms of relevant characteristics. Relevant characteristics refers to those variables that are somehow related to the dependent variable.
- Sample are only estimates
- The difference between the sample estimation and the true population is the **sampling error.**

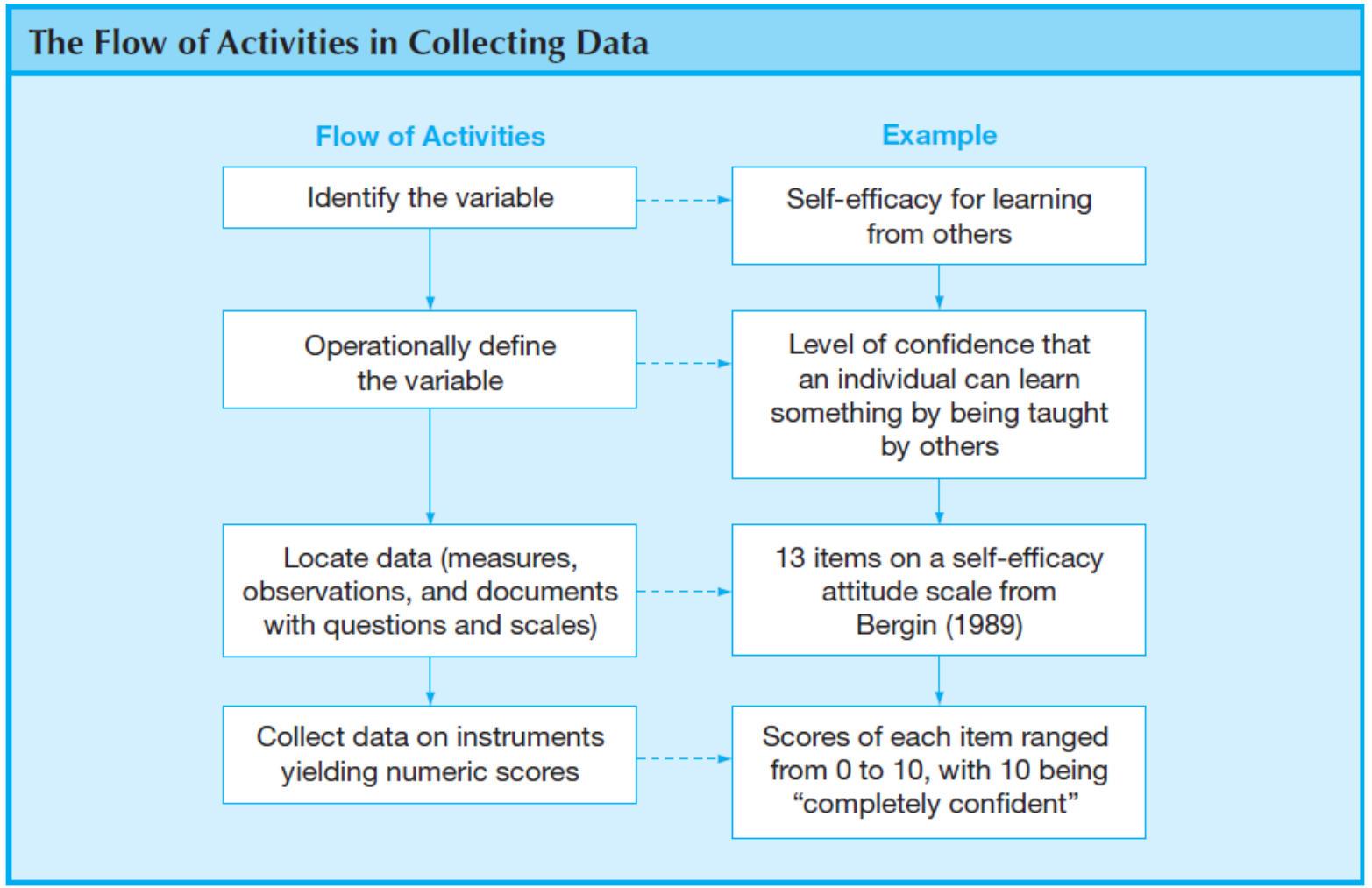
❖ Sample Size

- Sample should be as large as researcher can obtain with reasonable expenditure of time and energy
- Select a sufficient number of participants for the statistical tests that will be used
- Calculate the sample size using a sample size formula

❖ Procedure of Data Collection

1. Determining the participating
2. Obtaining permissions from individuals and organizations
3. Considering what types of information to collect from several sources
4. Locating and selecting instruments
5. Administering the data collection process

❖ Procedure of Data Collection



❖ Quantitative Sample Size

- Determine the size of the sample
- Select as large a sample as possible from the population
- The larger the sample, the less the potential error

❖ Quantitative Sample Size

- Obtain permission:
  - Institutions or organizations (e.g., school district)
  - Specific sites (e.g., the primary school)
  - A participant or group or participants
  - Parents of participants

❖ Types of Instrument

- Questionnaire
- Test





❖ Types of Instrument

➢ **Questionnaire** is a formalized set of questions for obtaining information from respondents

- Attitude
- Beliefs
- Behavior
- Attributes

❖ Types of Instrument

- **Questionnaire: Structure of Questions**
  - Open-ended question
  - Closed-ended question
- **Questionnaire: Types of Measurement Scales**
  - Likert Scale
  - Thurstone’s Types Scale
  - Osgood Scale/Semantic Differential Scale

❖ Types of Instrument

➢ Likert Scale or Summated Rating Scales

- Attitude scale which uses positive (+) and negative (-) equally weighted statements
- 3- or 4- or 5- or 7- point scale

❖ Types of Instrument

➢ Five-level Likert Scale: Agreement

Positive statement	Negative statement
1 = Strongly disagree	1 = Strongly agree
2 = Disagree	2 = Agree
3 = Neither agree nor disagree	3 = Neither agree nor disagree
4 = Agree	4 = Disagree
5 = Strongly agree	5 = Strongly disagree

❖ Types of Instrument

➢ Five-level Likert Scale

Frequency
1 = Always
2 = Very Often
3 = Sometimes
4 = Rarely
5 = Never

Quality
1 = Very Good
2 = Good
3 = Barely Acceptable
4 = Poor
5 = Very Poor

❖ Types of Instrument

➢ Osgood Scale or Semantic Differential Scale

- An attitude scale which measures differences in word meanings
- Measures people's reactions to stimulus words and concepts in terms of rating on bipolar scales defined with contrasting adjectives at each end
- Typically, a scale would consist of a minimum of five bipolar pairs

❖ Types of Instrument

➢ Osgood Scale or Semantic Differential Scale

- Check one blank between each pair indicating the opinion of the term on the scale
- Blanks are scored 1-7, providing a numerical score for the meaning of the term in each dimension

❖ Types of Instrument

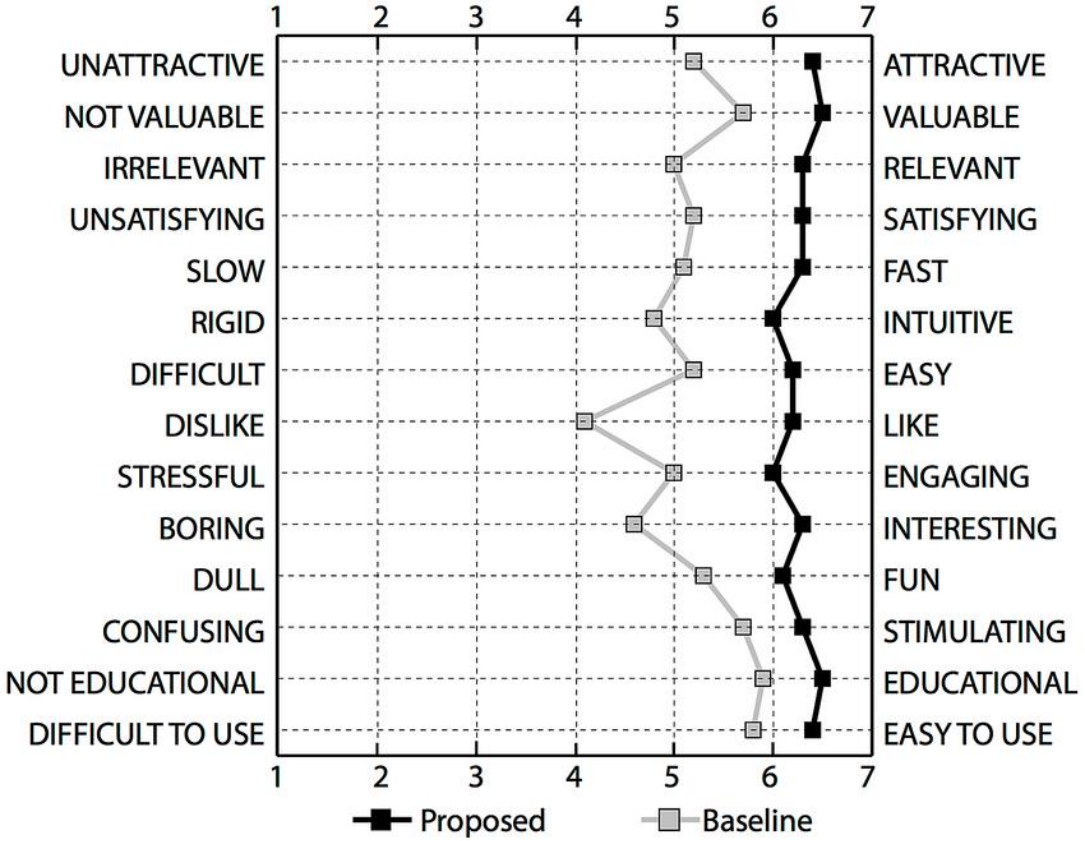
➢ Osgood Scale or Semantic Differential Scale

- A measure of leadership style (Fred Fiedler, 1964)

Pleasant	_ _ _ _ _ _ _ _ _	Unpleasant
Friendly	_ _ _ _ _ _ _ _ _	Unfriendly
Rejecting	_ _ _ _ _ _ _ _ _	Accepting
Tense	_ _ _ _ _ _ _ _ _	Relaxed
Distant	_ _ _ _ _ _ _ _ _	Close
Cold	_ _ _ _ _ _ _ _ _	Warm
Supportive	_ _ _ _ _ _ _ _ _	Hostile
Quarrelsome	_ _ _ _ _ _ _ _ _	Harmonious
Gloomy	_ _ _ _ _ _ _ _ _	Cheerful
Open	_ _ _ _ _ _ _ _ _	Guarded
Backbiting	_ _ _ _ _ _ _ _ _	Loyal
Untrustworthy	_ _ _ _ _ _ _ _ _	Trustworthy
Considerate	_ _ _ _ _ _ _ _ _	Inconsiderate
Nasty	_ _ _ _ _ _ _ _ _	Nice
Agreeable	_ _ _ _ _ _ _ _ _	Disagreeable
Insincere	_ _ _ _ _ _ _ _ _	Sincere
Kind	_ _ _ _ _ _ _ _ _	Unkind

❖ Types of Instrument

➢ Osgood Scale or Semantic Differential Scale



❖ Types of Instrument

- **Tests** are commonly used in research to measure personality, aptitude, achievement, and performance
- Three Type of Tests
  - Achievement Test
  - Intelligence and Aptitude Test
  - Personal – Social Test: is a questionnaire or other standardized instrument designed to reveal aspects of an individual's character or psychological makeup.

❖ Types of Instrument

➢ Achievement Test:

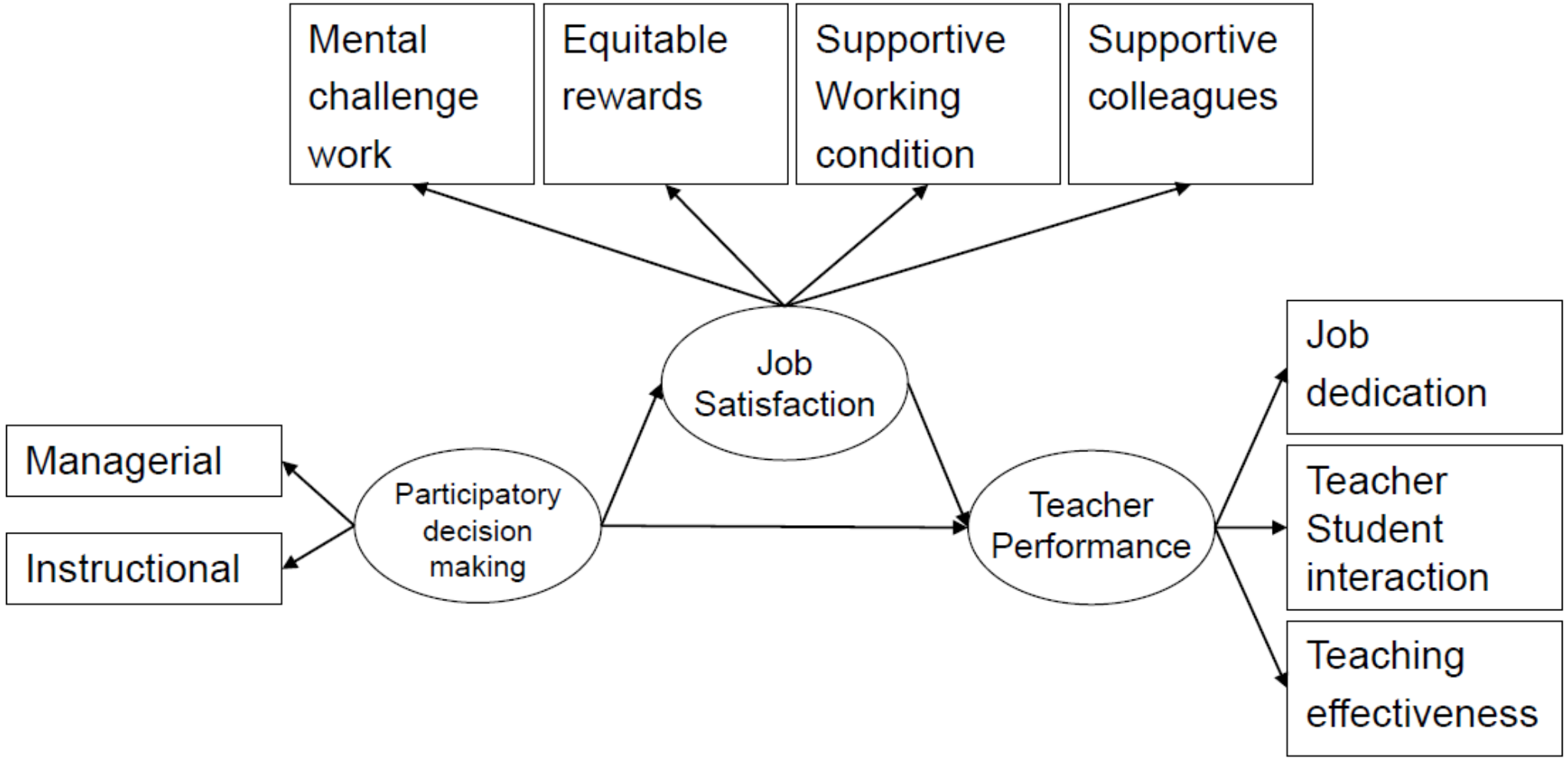
1. Oral Test
2. Paper-pencil Test: Essay type, Fixed-response type, True-false, Completion, Matching, Multiple choice
3. Performance Test

❖ Types of Instrument

➢ Intelligence and Aptitude Test:

1. Verbal factor, Word fluency factor
2. Number factor
3. Space factor
4. Memory factor
5. Perception factor
6. Reasoning factor
7. Individual Administered Test of General Aptitude, Group Test of General Aptitude

❖ Types of Instrument



❖ Types of Instrument

➢ Research Hypothesis

The casual model of teacher performance in Cambodia fit with the empirical data.

1. Teacher Performance will be affected directly by the effect of Participatory decision making and Job satisfaction
2. As mediated by Job satisfaction, participatory decision making effect will indirectly affect on teacher performance

❖ Types of Instrument

➢ Research Questionnaire

Section 1: Background information (check list: 6 items)

Section 2: Teacher performance (five-point Likert Scale: 48 items).

1. Participatory Decision Making: Decision Making Analysis (DIA) by Rice and Schneider (1994)
2. Job Satisfaction: Teacher Job Satisfaction Questionnaire (TJSQ) by Lester (1987)
3. Teacher Performance: Teacher Job Performance by Wang (2010)

Q&A